Bedfordview Primary School
TERM 4 Year End Test

Name:	Grade 5 : A B C D
Subject: Natural Science and Technology	
Examiner: Mrs C. Nash	Date: November 2023
Moderator: Mr J. Nash	Time Allocation: 1hour 30min Total: 60
INSTRUCTIONS 1. Read all the instructions and questions carefully. 2. Write neatly and legibly. 3. Write in full sentences unless otherwise instructed. 4. Look carefully at the mark allocation when giving your answers. 5. Answer in the spaces provided. This paper consists of 6 pages.	Parent's signature & comments: _____ _____ _____

1	2	3	4	5	6	7
0 – 29%	30 – 39%	40 – 49%	50 – 59%	60 – 69%	70 – 79%	80 – 100%

	60% Section A: Term 3 Energy & Electricity and Energy & Movement							40% Section B: Term 4 Planet Earth and Surface of the Earth			
Question	1	2	3	4	5	6	7	1	2	**Total**	**%**
Allocated mark	**7**	**3**	**13**	**5**	**3**	**7**	**3**	**11**	**8**	**60**	
Learner's mark											
Moderated mark											

Content: **Energy and Electricity**: cells and batteries; mains electricity. **Energy and Movement**: wheel and axle; elastics and springs. **Planet Earth**: the earth moves and **Surface of the Earth**
Skills: Understanding and interpreting data on storing of energy, circuits, electricity and transferring of energy. Identifying objects that are elastic. Understanding stored energy and movement of energy

SECTION A – Energy and Electricity & Energy and Movement **/41**

Question 1: Definitions (Knowledge recall) **/7**

Match the words in column A with their meaning in column B.

Column A	**Column B**
A. Components	1. Things you do to stop something dangerous happening.
B. Rotation	2. The movement of an object in space around another object.
C. Precautions	3. Groups of related parts that work together as a whole for a particular purpose.
D. Systems	4. The parts that make up an electric circuit.
E. Revolution	5. Kinetic energy.
F. Compress	6. The movement of an object around itself.
G. Movement	7. To press or squash together.

A-	B-	C-	D-	E-	F-	G-

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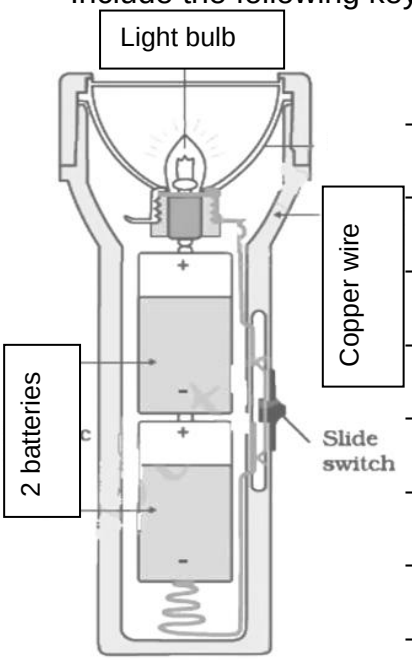
<u>Questions 2: Cells and batteries</u> (Data interpretation)

2.1 Describe how the electricity would flow through a torch.

Use the diagram with labels to guide you.

Include the following key words: **battery, copper wire, light bulb, switch** **/3**



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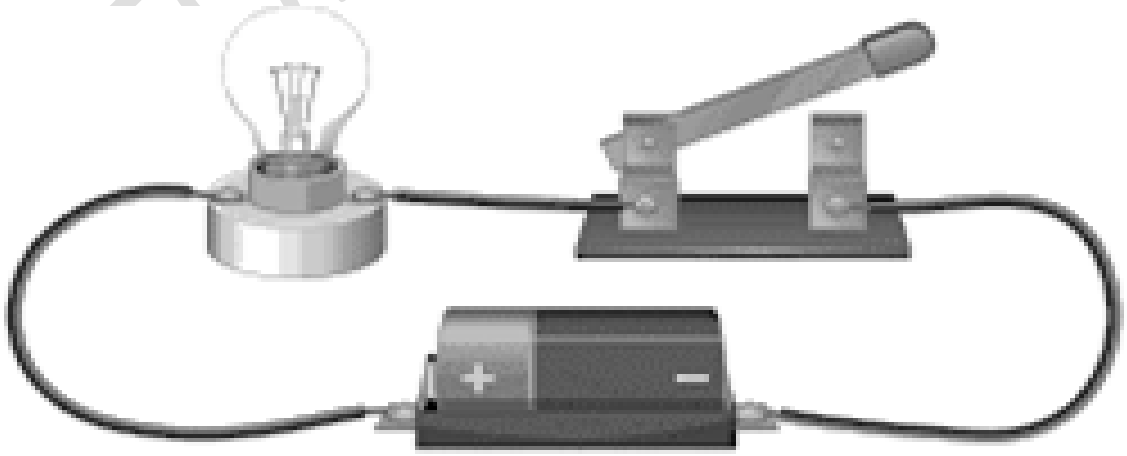
<u>Questions 3: Electrical Circuit</u>

Study the electrical circuit below and answer the questions that follow.

3.1 **Explain the function** of each component in the following electrical circuit: **/4**



Battery:_____

Switch:_____

Light bulb:_____

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Copper wire:_____

3.2 **Draw and label** a closed **electrical circuit** with all the necessary **components** using the correct symbols in the space provided. **/8**

Diagram	**Labels**

3.3 Predict what would happen if the copper wires are not properly connected to the electrical components? **/1**

Explain your reasoning.

_____

_____

Question 4: **/5**

Draw a circle around the letter (A – D) to select the correct answer.

1.1 For a candle to keep on burning, the flame needs …

a) Oxygen
b) Carbon dioxide
c) Oxygen and Carbon dioxide
d) None of the above **/1**

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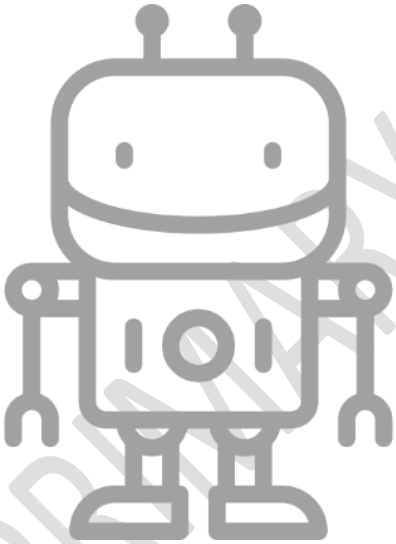
1.2 Stored energy can be changed into… energy in a toy car.

a) Light
b) Kinetic
c) Sound
d) Heat **/1**



1.3 An example of a useful output energy is…

a) Heat
b) Noise
c) Smoke
d) None of the above **/1**

1.4 To prevent a fire from spreading we should…

a) Always play with matches and lighters
b) Never leave a fire unattended at a camp site
c) Place wet clothes over a heater to dry
d) Stay indoors and make no attempt to get out **/1**

1.5 An important example of an everyday fuel is…

a) Petrol
b) Water
c) Oxygen
d) Food **/1**

Question 5: **/3**

Select the correct answer to make the following statements true.

1.1 A radio changes electrical energy into _____energy **/1**

a) Light
b) Heat
c) Movement
d) Sound

1.2 South Africa's main power station uses _____ **/1**

a) Water
b) Wind
c) Sun
d) Coal

1.3 The input energy for an electrical lightbulb is_____ **/1**

a) Light

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b) Heat
c) Electricity
d) Stored

Question 6: Mains electricity **/7**

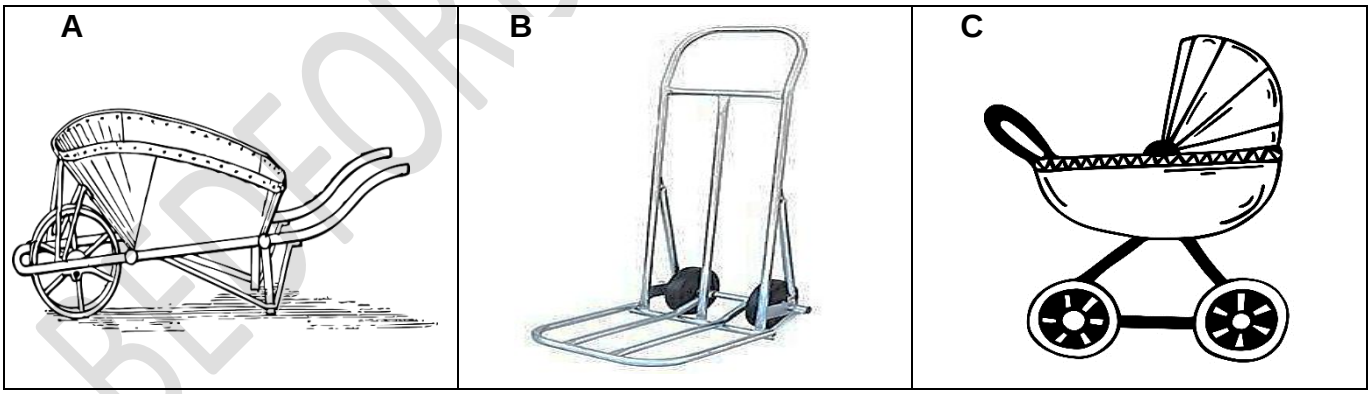
Place the following statements in their correct order:

How electricity flows from power stations to our homes, schools, and offices.	Order
1. Substations reduce high-voltage electricity to a lower voltage, which can be used in homes and offices.	
2. An electrical appliance draws the energy needed to work from the mains supply.	
3. Powerlines carry electricity from the power stations to substations in areas where people live and work.	
4. Wall sockets built into the walls of a building provide places for people to safely plug into the mains supply.	
5. When plugged into the wall socket, an appliance is connected to the mains electrical supply.	
6. Power stations burn fuels such as coal to generate electricity.	
7. Electricity boxes at home control the supply of electricity to the building.	

Question 7: Systems for moving things (Distinguish)

Examine the following images and answer the questions that follow.

7.1 How would the following system make a person's life easier. **/3**



a) _____

_____

b) _____

_____

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c) _____

_____

SECTION B – Planet Earth **/ 19**

Question 1:

State whether the following sentences are true or false. **/11**
If false, rewrite the statement to make it true.

a) The Earth's axis is vertical. _____

_____

b) The whole Earth can receive light at the same time. _____

_____

c) The side of the Earth that receives light is experiencing night. _____

_____

d) The Earth takes 24 hours to complete one rotation on its axis. _____

_____

e) The Sun takes 365 ¼ days to revolve around the Earth. _____

_____

f) The shape of the Earth's orbit is square. _____

_____

Question 2: **/8**

2.1 Write a full sentence explaining **where soil comes from.** **/1**

_____

_____

2.2 Write a full sentence explaining **what soil is made up of.** **/4**

_____

_____

2.3 Complete the following table by filling in the missing information. **/3**

	Clay	**Loam**	**Sand**
Size of particles	Most of the particles in clayey soil are small		Most of the particles in sandy soil are quite large

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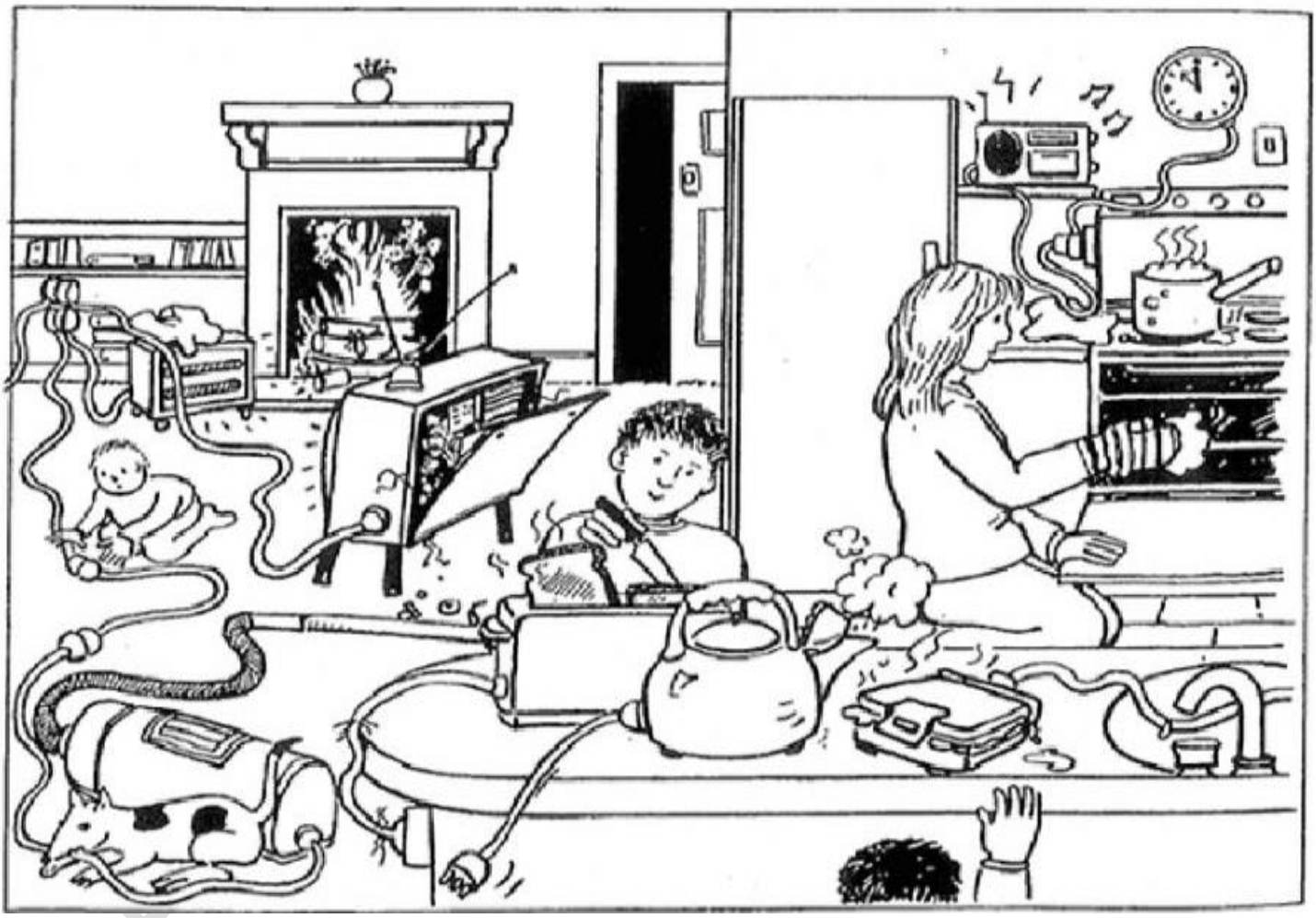
Drainage	Clayey soil holds a lot of water	Loam soil lets some water pass through	
Nutrients		Loam contains the right amount of nutrients	Sandy soil contains very few nutrients

Extension Activity

To be completed after you have checked **all your answers** and only if you have finished.

Spot the Electrical Hazards